

Executive Summary & Central Conclusion

Project: Integrated Multi-Omics Target Validation for In Vivo Type 2 Diabetes (T2D) Study
Framework: Adapted from *Xu et al., Phytomedicine 154 (2026) 158050*
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1. Executive Summary

This study establishes a rigorous, end-to-end multi-omics target nomination and validation pipeline for Type 2 Diabetes (T2D). By integrating **human liver bulk transcriptomics (WGCNA)**, **KEGG pathway topology (KGML)**, **mouse single-cell RNA sequencing (liver & kidney atlases)**, **machine learning consensus modeling**, and **Mendelian Randomization (MR)**, we identify the primary mechanistic regulatory axis driving diabetic tissue stress and nominate prioritized targets for in vivo pharmacological validation.

THE 5-STAGE ANALYTIC FUNNEL

[Stage 1: Human Bulk]

1,648 DEGs n WGCNA Key Modules n Hyperglycemia Set

(8 Core Candidates)

[Stage 2: Pathway/KGML]

Top KEGG Map: AMPK Signaling (hsa04152; ORA q = 5.8e-6)

(Candidate-Anchored Axis)

[Stage 3: scRNA Atlas]

Cross-Species Cell Confirmation (Liver Cholang. & Kidney EC)

(Spatial Co-Localization Gate)

[Stage 4: ML Consensus]

Feature Ranking (LASSO / Random Forest / SVM Consensus)

(Multi-Variable Predictability)

[Stage 5: Human Genetics]

cis-eQTL Mendelian Randomization (Outcome: 74k T2D Cases)

[DELIVERABLE]

Tiered Target Portfolio for In Vivo Validation (GV1 vs Met)

2. Central Biological Conclusion

Suppression of the AMPK Signaling Axis Output

The central finding across human and rodent diabetic datasets is the **functional suppression of AMPK pathway activity** under chronic hyperglycemic/insulin-resistant stress:

Suppressed AMPK Output:

* PPARGC1A (PGC-1α) ↓ → Loss of mitochondrial biogenesis and hepatic metabolic control.

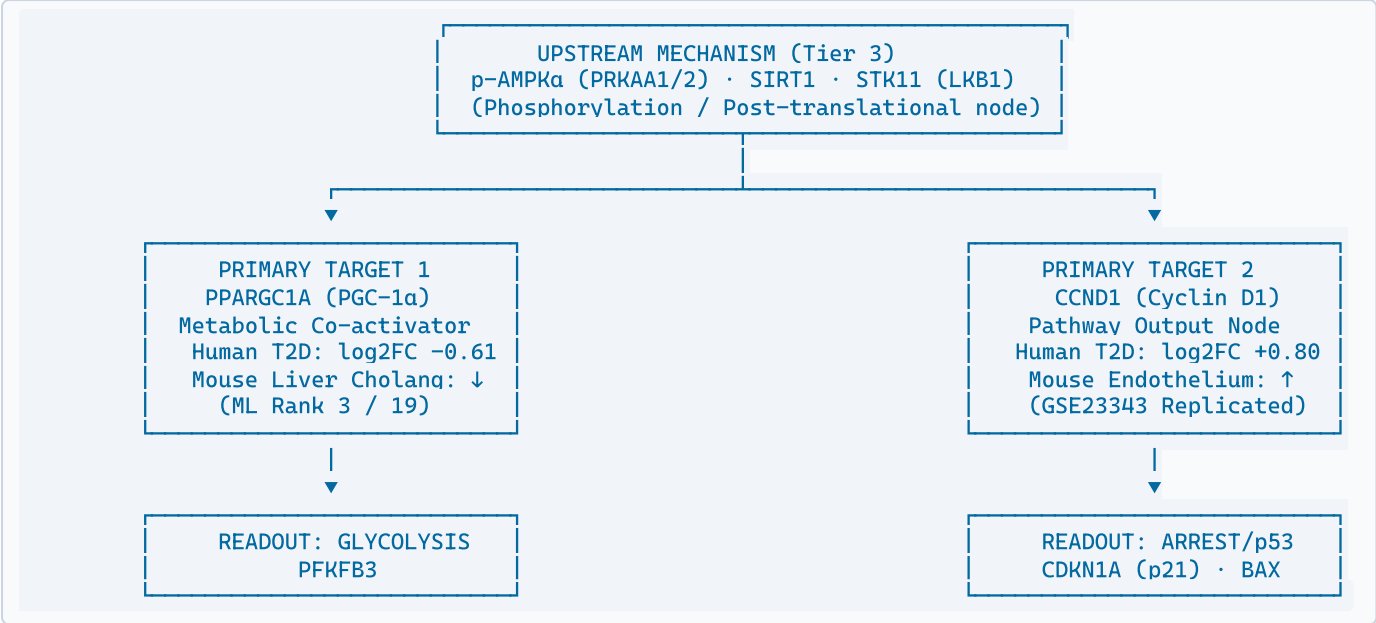
* CCND1 (Cyclin D1) ↑ → De-repressed cellular stress and aberrant proliferation signal.

Key Mechanistic Insights:

1. **Post-Translational Control Signature:**
The human KEGG network displays discordant edge dynamics: the AMPK kinase subunit (*PRKAA2*) is transcriptionally maintained while its activated downstream targets fall. AMPK activity is governed by **Thr172 phosphorylation** and cofactor ratios, not mRNA abundance.
2. **Epithelial & Endothelial Compartmentalization:**
3. In **liver**, metabolic suppression (*Ppargc1a* downregulation) localizes specifically to epithelial **cholangiocytes** (per-mouse $P = 0.011$, detection dropping from 31.7% to 11.4%).
4. In **kidney**, stress/cell-cycle de-repression (*Ccnd1* upregulation) localizes to **microvascular endothelial cells** (pseudobulk $P = 1.2 \times 10^{-6}$).
5. **Cross-Species Concordance:**
The directional shifts of both primary anchors (*PPARGC1A* down, *CCND1* up) replicate across human clinical cohorts and mouse diabetic models.

3. Core Target Tiers & Evidence Hierarchy

Targets are classified mechanically into prioritized tiers based on evidence convergence across all five analytical stages:



Target Classification Table

Target	Tier	Role & Mechanism	Key Supporting Evidence	Planned In Vivo Assay
PPARGC1A <i>(PGC-1α)</i>	Tier 1 (Primary)	Master transcriptional coactivator of mitochondrial respiration & gluconeogenesis.	- Human liver: log₂FC = -0.61, <i>P</i> = 6.0 × 10⁻⁴ - Mouse cholangiocytes: <i>P</i> = 0.011, cell BH <i>P</i> = 2.0 × 10⁻⁸ - ML Consensus: Rank 3 / 19	- RT-qPCR & Western blot (liver) - IF co-staining: PGC-1α × CK19
CCND1 <i>(Cyclin D1)</i>	Tier 1 (Primary)	De-repressed downstream output of AMPK → HuR/ELAVL1 axis.	- Human liver: log₂FC = +0.80, <i>P</i> = 7.5 × 10⁻⁴ - Replicated in human cohort GSE23343 (<i>P</i> = 0.016, AUC = 0.81) - Mouse kidney endothelium: <i>P</i> = 1.2 × 10⁻⁶	- RT-qPCR & Western blot (kidney & liver) - IF co-staining: Cyclin D1 × CD31
CDKN1A <i>(p21)</i>	Tier 2 (Readout)	Cell-cycle arrest / senescence effector downstream of metabolic stress.	- Human liver DEG: log₂FC = +0.96, FDR = 0.016 - Elevated in mouse cholangiocytes (FDR = 0.009) & kidney endothelium (FDR = 0.008)	- RT-qPCR & Western blot - IF co-staining with CD31 in kidney
PFKFB3	Tier 2 (Readout)	Inducible glycolytic regulator connected to AMPK.	- Human liver: FDR = 0.042 - Mouse liver LSEC (<i>P</i> = 0.049) & kidney LoH (BH <i>P</i> = 0.03)	RT-qPCR & Western blot (liver & kidney)
PRKAA1/2 <i>(AMPKα)</i>	Tier 3 (Master Node)	Central metabolic sensor and upstream serine/threonine kinase.	Upstream driver on KEGG hsa04152; acts via Thr172 phosphorylation.	- Western blot for p-AMPKα (Thr172) / total AMPKα - Metformin positive control

4. Multi-Modal Evidence Matrix

Candidate Gene	Human Bulk DE	WGCNA Module	KEGG AMPK Map	Mouse scRNA Validation	Human Replication	ML Consensus	Causal MR (GWAS)	Target Tier
PPARGC1A	Down ($P = 6.0 \times 10^{-4}$)	Key Module	Direct Step (+1)	Down (Cholang. $P = 0.011$)	Concordant	Rank 3	Null ($P = 0.16$)	Tier 1
CCND1	Up ($P = 7.5 \times 10^{-4}$)	Key Module	2 Steps via HuR (-1)	Up (Kidney EC $P = 1.2 \times 10^{-6}$)	Replicated ($P = 0.016$)	Rank 14	Null ($P = 0.55$)	Tier 1
CDKN1A	Up (FDR = 0.016)	Key Module	p53 / Arrest Map	Up (Cholang. & Kidney EC)	Concordant	—	Null	Tier 2
PFKFB3	Down (FDR = 0.042)	Turquoise	Direct Step (+1)	Down (LSEC $P = 0.049$)	Concordant	—	Null	Tier 2
IRS2	Down ($P = 0.011$)	Key Module	Input Node	Minor Cells only	Concordant	Rank 6	Null	Tier 2
IGF1	Down ($P = 0.021$)	Key Module	Input Node	Sparse (Hepatocyte gap)	Concordant	Rank 8	Null	Tier 2b
PRKAA2	Up ($P = 0.017$)	Network Node	Master Kinase	Post-translational	Concordant	—	Null	Tier 3
SREBF1	Down ($P = 0.041$)	Network Node	Lipogenic Output	Inverted (STZ model)	—	—	Causal (Blood, $P = 9.0 \times 10^{-9}$)	Exploratory
VSNL1 / LZTFL1	DE (FDR < 0.05)	Key Module	Off-Map	Not DE in mouse	Weak	Rank 1 & 2	Null	Non-Target

5. In Vivo Experimental Validation & Pharmacological Rescue Plan

The findings directly define the experimental design and endpoints for the in vivo animal validation study:

IN VIVO VALIDATION DESIGN (GV1 vs MET)	
[Groups]	1. Normal Control (Saline) 2. Ob-T2D Disease Model (High-Fat Diet / STZ) 3. Ob-T2D + GV1 Low Dose 4. Ob-T2D + GV1 Medium Dose 5. Ob-T2D + GV1 High Dose 6. Ob-T2D + Metformin (Positive Control, 200 mg/kg)
[Primary]	1. Glycemic & Metabolic Efficacy: OGTT, Fasting Glucose, HbA1c, HOMA-IR
[Molecular]	2. Hepatic AMPK Restoration: Western blot p-AMPKα (Thr172) / total 3. Mitochondrial Rescue: PGC-1α expression (RT-qPCR & Western blot) 4. Proliferation / Stress Normalization: Cyclin D1 & p21 (qPCR & WB)
[Histology]	5. Immunofluorescence: PGC-1α × CK19 (liver); Cyclin D1 × CD31 (kidney)

Hypothesized Therapeutic Mechanism of Action:

Administration of the active formulation (GV1) is hypothesized to mirror the therapeutic action of **Metformin** by:

1. Re-activating hepatic **AMPK phosphorylation (Thr172)**.
2. Restoring downstream **PGC-1α** expression and mitochondrial function.
3. Normalizing aberrant **Cyclin D1** and **p21** levels, resolving chronic tissue stress in liver and kidney.